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Urban food policies and their influence on the development of Territorial Short Food Supply Chains: The case of cities in Colombia and Spain

Liliana Reina-Usuga [a,*], Carlos Parra-López [b], Tomás de Haro-Giménez [a]

[a] University of Córdoba. Department of Economics, Sociology and Agricultural Policy, Campus de Rabanales, P.O. Box, 14071 Córdoba, Spain
[b] Institute of Agricultural and Fisheries Research and Training (IFAPA), Department of Agrifood System Economics, Centro 'Camino de Purchil', P.O. Box 2027, 18080 Granada, Spain

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ABSTRACT

Urban food policies (UFPs) have emerged as a key mechanism to drive the transition towards more sustainable food systems, with cities as the appropriate scale for their implementation. The aim of the paper is analysis of the boost that UFPs have given to one of the most widespread forms of Alternative Food Networks, these are Territorial Short Food Supply Chains (TSFSCs). For this, the territorial factors that strengthen TSFSCs in the cities of Bogotá (Colombia) and Córdoba (Spain), one being UFP, are prioritised using a multi-criteria model based on the Analytic Network Process. Based on this prioritisation, the way the UFPs interrelate with the other prioritised territorial factors in both cities is identified. Consequently, the influence of UFP implementation on social capital of TSFSCs is analysed through Social Network Analysis. The results indicate that there are four types of priority territorial sub-factors on which UFPs should have an impact in order to promote TSFSCs in the two cities. The first group is classified as means to achieve the other sub-factors. The other sub-factors are classified as ends. The main means are UFPs and partnership linkages. The territorial sub-factors included in the UFP are in the design and implementation phases of the policy cycle. Furthermore, UFPs foster bridging social capital, with actors connecting unconnected nodes. Finally, some of the reflections on the implications of UAPs indicate that the promotion of participatory governance mechanisms involving civil society is an important element to include in UFPs, given their influence on strengthening collective action and social capital in cities. This shows that a territorial approach in UFPs can have greater results in policy implementation. In this way the paper contributes, at the theoretical and empirical level, to recent debates on UFP approaches and the integration of key factors for food transformation in cities.

1. Introduction

In the last decades, different economic, social, environmental and institutional factors have shaped a scenario for food, in which a Global Agrifood System (GAS) has imposed itself over time. GAS is formed by transnational corporations, large agri-food companies and corporate retailers (Ilbery et al., 2004; Whatmore et al., 2003). In this system, food provisioning is characterised by three mutually reinforcing processes: 1) disconnecting (farmers and consumers); 2) disembedding (food and territory); and 3) 'disentwining' (farmers and suppliers) (Ilbery and Maye, 2015).

However, the design and diffusion of different, more sustainable Food Systems (FS) has become a high priority issue both at the scientific level and the international political agenda. The challenges of achieving more sustainable FS have involved a number of transitions in the public policy sphere, implying a change from the classical sectoral policy approach, which sometimes is fragmented and hard to implement (Sonnino et al., 2019; Lang and Barling, 2012) to a new integrated and systemic approach (Galli et al., 2020; IPES Food, 2019). In this regard, there is considerable debate in the academic literature over land use and food security and the range of possible futures for the FS (Mora et al., 2020). Some studies have identified urbanization, food consumption

Abbreviations: AFN, Alternative Food Networks; CBMPC, Coordinating Board Milan Pact in Córdoba; FS, Food Systems; FT, Food Territory; GAS, Global Agrifood System; MPFSSB, Master Plan for Food Supply and Security in Bogotá; MUFPP, Milan Urban Food Policy Pact; SDG, Sustainable Development Goals; TSFSC, Territorial Short Food Supply Chain; UFP, Urban Food Policies.

* Corresponding author.
E-mail address: Z52reusm@uco.es (L. Reina-Usuga).

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and diets as key drivers of future land use (Fritsche et al., 2015). For example, there are currently competing forms of land use that deepen the dependence of urban food systems on GAS, leading to a reduction in urban food production and urban food security (Olsson et al., 2016). This situation is caused by the lack of a specific policy for food production in peri-urban and urban regions, as there is for land use for housing areas and other building activities, recreational activities, as well as for the conservation of biodiversity and cultural values (Morgan, 2015).

The goals of new food policies should include coordinating and aligning actions across different policy domains and levels of government (IPES Food, 2019), promoting the participation of all actors in the food territory (FT) (Galli et al., 2020), as well as addressing current challenges and priorities such as the Sustainable Development Goals (SDGs). In the latter area, Alternative Food Networks (AFN) emerge as an innovative vehicle for creating a more sustainable food paradigm, which aims to simultaneously respond to demands of food security and sustainability (Marsden and Morley, 2014).

Recent examples, both empirical and theoretical, have positioned cities as the optimal scale for Urban Food Policy (UFP) innovation (Brand et al., 2019; Moragues-Faus and Sonnino, 2018; Moschitz, 2018; Sonnino, 2016; Sonnino et al., 2019), as they foster understanding of local FS dynamics and dialogue between civil society and local governments. This new approach also demands a broad reorganisation of the strategic tools in UFP to drive 'bottom-up' processes (Galli et al., 2020; Reina-Usuga et al., 2020).

However, the transition towards an integrated approach in public policies is not achieving the expected results, neither at the level of integration nor implementation (Galli et al., 2020; Pe'er et al., 2019). For example, in the area of sustainable food systems, despite the fact that many cities have adopted UFPs, which include AFN; and many others have joined global initiatives such as the Milan Urban Food Policy Pact (MUFPP), which is first of all a Pact signed by city's mayors (https://www.milanurbanfoodpolicypact.org/wp-content/uploads/2020/12/Milan-Urban-Food-Policy-Pact-EN.pdf.); improvements in food security, effective stakeholder participation and legitimacy of processes are not having the expected impacts (Blay-Palmer et al., 2020; Brand et al., 2019; Doernberg et al., 2019; Moschitz, 2018; Sonnino et al., 2019). Sonnino et al., 2019 highlight that an increasing number of municipalities are experimenting without financial support and without an overarching policy framework. This suggests the need for more tools and design elements for UFPs to truly achieve their purpose.

Although the approaches proposed for UFPs allude to an integrative focus, they have not yet embraced a territorial perspective considering all the elements that shape and define the territory. Therefore, the key factors that could explain the success in promoting sustainable urban food systems, as well as the most appropriate mechanisms and UFPs to foster them, have not yet been sufficiently identified. This manuscript presents an analysis of the boost that UFPs have given to one of the most widespread forms of AFN, the Territorial Short Food Supply Chains (TSFSCs), through a comparative study of two cities with different geographical, demographic and UFP integration features. The specific objectives are: 1) to prioritise the territorial factors that strengthen TSFSCs in the cities of Bogotá (Colombia) and Córdoba (Spain), one of these factors being UFP; 2) to identify how the UFPs interrelate with the other prioritised territorial factors in both cities, and 3) to analyse the influence of UFP implementation on the development of TSFSCs, through social capital analysis, in each city. In the first section of this paper, we show a framework over territorial approach based on Latin[1] literature of territorial agrifood system and AFN, which brings to the concept of Territorial Short Food Supply Chain (TSFSC). In the second section, we apply this approach to two case studies – Bogotá (Colombia) and Córdoba (Spain) - and conclude with a discussion of how UFP can foster TSFSCs as a sustainable vehicle for the FS.

2. Territorial approach: A theoretical framework

2.1. From space to territory

The approach to the food issue from different geographies highlights the singularity of the narratives that emerge in each place and that shape food strategies (Sonnino, 2016), which are limited or enhanced by the territorial factors specific to the place where they are developed. Here, the notion of territory goes beyond the reductionist and static idea of a mere given physical space, provider of natural and material resources, to be conceived as a process of social construction determined by local actors and their interactions to enhance the value of territorial resources (Pecqueur, 2001; Sánchez-Zamora et al., 2014). Thus, territory can be understood as a complex system composed by: i) the topographic metric space and its territorial resources (Lamara, 2009; Moine, 2006).; ii) territorial agents, who are the actors that make territory (Campagne and Pecqueur, 2014), and iii) the institutional agreements, with the potential for generating social capital (Lamara, 2009; Moine, 2006; Woolcock and Narayan, 2000; . The specific combination of the three elements mentioned above, lead to the territorial factors or also known as territorial capitals (Emery and Flora, 2006). Thus, the interaction of territorial factors influences the processes of territorial organisation (D'Aquino, 2002) and is closely linked to the study of collective action (Moyano, 2005) and governance (Candel, 2014).

The territorial approach has extensive implications for policies linked to rural areas (FAO/OCDE/FNUDC, 2016), underpinned by the theoretical framework of the endogenous development. For instante, the Common Agricultural Policy (CAP) and in particular its 'second pillar' – the Rural Development Regulation -, includes it. The dimension associated to local development highlights the use of endogenous resources available in the territory, to stimulate and promote economic growth (Vázquez Barquero, 2005) and to overcome the challenges of food security (FAO/OCDE/FNUDC, 2016). This approach acknowledges the heterogeneity of the features of the territories and communities, and encourages the representation of local priorities in the decision-making processes (Ramos and Garrido, 2014). Therefore, the territorial approach could be a useful complement to achieve integrated UFPs and increase their capacity to transform urban FS.

2.2. Territory and Alternative Food Networks

In the academic literature, two theoretical perspectives are differentiated to tackle the concept of the AFN: the first focuses on the role of production and the possible contribution of this type of network to sustainable rural development (Goodman et al., 2012; Ilbery and Maye, 2015; Mundler and Laughrea, 2016; Renting et al., 2012); and the second on the active role of urban consumption from the perspective of sustainable consumption as social movements, or as attempts to transform FS structures (Lamine et al., 2012; Moragues-Faus, 2017; Renting et al., 2012; Sonnino, 2016; Sonnino et al., 2014). However, public policy support to AFN has been mainly based on the first production perspective and has been included in rural development programmes (Galli et al., 2020; Pothukuchi and Kaufman, 1999). Only recently, many cities and regions have made up their UFPs. Some of them have included AFN not only in a production perspective, but also in a consumption promotion perspective. However, there are many other cities that, without having a specific UFP, have joined international initiatives such as the MUFPP, as a first step towards a future design of such a UFP (Candel and Pereira, 2017; Doernberg et al., 2019; Filippini et al., 2019; Sibbing et al., 2019; Sonnino, 2016). Although the emergence of UFPs is promising (Sibbing et al., 2019) and many have an orientation towards AFN, there is still a great uncertainty about how UFPs are designed, how

[1] *Latin* refers here to countries where Romance languages are spoken (mainly French, Italian, Spanish and Portuguese) both in southern Europe and Latin America.

they are applied in normative and in practice, and who should foster and implement them (Cretella, 2019).

The new relationships between production and consumption, developed within AFNs, are closely linked to the territory where they are developed, as this gives them a unique feature (González et al., 2012). AFNs are broader than localized agrifood systems (SYAL), as the former are not only limited to specialty products (such wine, olive, or cheese), but also include a larger diversity of ordinary food products embedded in the territory (Lamine et al., 2019). However, AFNs shares some characteristics with SYAL, such as that a wide variety of plants, indigenous breeds and traditional knowledge accumulated over time flow through them, contributing to the preservation of the cultural heritage of the territories (Sanz-Cañada and Muchnik, 2016).

2.3. Territorial Short Food Supply Chains

One of the prominent types of collective action in the agrifood sector within the AFN, is the Territorial Short Food Supply Chain (TSFSC). It is defined as forms of rural-urban interaction of different food actors, with their knowledge and interests, who converge to develop collective visions around a specific food territory, with objectives related to sustainable values, and with a clear focus on agrifood governance (Reina-Usuga et al., 2018). In this context, food territory is understood as the social construction (resources, actors and agreements) around food issues, in particular linked to guaranteeing the right to food and food sovereignty.

According to Reina-Usuga, 2018, TSFSCs can have two distribution channels (without being mutually exclusive): one through direct sales (such as farmers' markets), and another with a single intermediary between production and consumption (such as local shops). They are characterized by: (i) a strong social base that supports them and facilitates their development; this indicates that TSFSCs are supported by the social capital formed among the food actors involved and is an essential element for their emergence and growth; (ii) the participation of small farmers and retailers; (iii) building trust and quality standards among the linked actors (institutional agreements), and iv) linkages with alternative/diverse economies (e.g. social and solidarity economy, economy for the common good, etc.). Coordination mechanisms of the SFSCs are mainly settled in the social and market sphere, and foster 'spaces of deliberation' that bring together a wide variety of actors with different interests, e.g. civil society, private agents, and local government, promoting processes of learning and collective action, thus configuring a reflexive and networked governance. These two governance approaches (reflexive and networked) operate in tandem, with important complementary and synergistic effects that foster food democracy (Reina-Usuga et al., 2020). Thus, TSFSCs replace competing market relationships with trust and collaboration, while renewing the chain of command hierarchies with a more horizontal structure (Bevir, 2012). This aspect acquires importance given the consolidation and concentration of power that has been identified in the GAS over the last 20 years, both in the supply of inputs and in the distribution of food (McMichael, 2009; IPES-Food, 2017). These unequal relationships in food chains are recognised as one of the greatest vulnerabilities to food security (Moragues-Faus et al., 2017).

3. Material and methods

3.1. Case studies

The cities of Bogotá (Colombia) and Córdoba (Spain) were selected due to the growth experienced in recent years by the TSFSCs and the different nature of each UFP in each city, the former being linked to a food supply plan and the latter to an international framework policy. Furthermore, the international scope of the research is based on the representativeness of both cities of two very different contexts. Bogotá has the typical characteristics of alternative Latin American AFNs, such as: peasant agriculture (Ranaboldo and Arosio, 2014), agroecological practices (Van der Ploeg et al., 2010), local trust labels (van der Ploeg et al., 2012), and international cooperation assistance (through NGOs) (Reina-Usuga et al., 2020). On the other hand, Córdoba represents European AFNs, characterized by certified organic agriculture (González et al., 2012), high consumer involvement (Moragues-Faus, 2017; Renting et al., 2012) land and transport property rights (Fonte and Cucco, 2017), and support from the Common Agricultural Policy (CAP) (Moragues-Faus et al., 2017).

3.1.1. Bogotá (Colombia)

The city of Bogotá, with a population of 8,181,047 inhabitants (DANE, 2018) concentrates the commercialization of fruit and vegetable products in the Central de Abastos de Bogotá (Corabastos), the central supply market of the city. It is estimated that local shops, market places and specialised shops (which supply 60–65% of final consumption), are supplied with almost 100% of the perishable products by Corabastos (Galindo, 2015). In Bogotá, about 65% of the food consumed is supplied by peasant economies (Gutierréz, 2016) and there is evidence of a high level of intermediation in the supply food chain (Parrado and Molina, 2014).

Regarding UFPs, these have been promoted in the cities of Bogotá and Medellín, being the former pioneer in this type of initiative in Latin America (Gutierréz, 2016). Specifically, the Master Plan for Food Supply and Security in Bogotá (MPFSSB), established in 2006 as a public policy instrument to regulate food supply in the city. The MPFSSB is part of the Land Use Plan (POT - by its acronym in Spanish), which incorporates actions to physically organize the territory and the logistical equipment (physical connections) have been privileged. Moreover, this Plan has been carried out actions at the supply (agroredes) and demand (nutriredes), as well as boosting of the peasant economy through farmer markets. However, to date, the Plan has not been completely carried out and there are failures in the components that have been implemented (Bayona Romero, 2013; Parrado Barbosa et al., 2011; Ramírez, 2010). Despite this, the Mercados Campesinos program (which began in 2004 and became part of the MPFSSB in 2007), based on TSFSCs that support the peasant economy, has become a form of alternative food supply in Bogotá. At the same time, other TSFSC initiatives linked to agro-ecological issues have emerged, driven mainly by NGOs and universities (Reina-Usuga et al., 2020). Subsequently, in 2014, the Special Administrative and Planning Region (RAPE - by its acronym in Spanish) was created with the purpose of promoting and articulating regional plans for social, economic, and environmental development in its territories. Its scope and competencies in the food sector are not entirely definitive, but this entity coordinates the current Mercados Campesinos program. Finally, there is the Colombian Peace Agreement signed in 2016, where food security is a cornerstone, and the promotion of local and territorial markets is an important pillar of the proposed comprehensive rural reform. Although the agreement has not been fully implemented either.

3.1.2. Córdoba (Spain)

The city of Córdoba has 328,718 inhabitants (Ayuntamiento de Córdoba, 2019) and is the third largest and most populated city of Andalusia. It belongs to the province of the same name, in which an eminently agricultural vocation stands out, with special importance given to the cultivation of olive groves and livestock linked to the dehesa, a multifunctional, agrosilvopastoral system and cultural landscape of southern and central Spain and southern Portugal. Mercacordoba is the central supply market of the city and provides 88% of the supply of perishable produce to local shops (Gallar and Vara, 2017). The province has seen an increase in organic production in recent years (Junta, 2012; MAPAMA, 2015), and throughout Andalusia there has been an increase in the number of wholesalers and small retailers specialising of organic product (MAPAMA, 2015).

As for AFNs, in 1994 the first cooperative of consumers of ecological

products was registered in Córdoba, but it is not until the decade of 2000 when these types of initiatives become increasingly important (Sanabria, 2012) and the city experienced a proliferation of different forms of self-supply and TSFSCs.

Regarding UFPs, as a European Union country, Spain is subject at the macro level to the policies and programmes authorised by the European Commission. In the agricultural and food sphere, the CAP and the regulations and directives on food issued by the Council and the European Parliament stand out, the latter known as the health and hygiene package. Within the framework of the hygiene package, Member States may apply flexibility criteria according to the country's own characteristics. In this framework, Spain has promoted a few decrees to promote TSFSCs. Nevertheless, cities can carry out their own UFP, as long as they are aligned with the EU food safety standards. Thus, Córdoba joined the MUFPP in 2016. This policy agenda focuses on the promotion of sustainable, inclusive, and resilient FS, with the TSFSCs as a priority line. This policy has led to the creation of a space for deliberation and decision-making by civil society organisations and the local public administration, known as the Coordinating Board Milan Pact in Córdoba (CBMPC). It is worth noting that this policy is not institutionalised as a normative framework, nor is it embedded in a land use or food security plan, as is the case in Bogotá.

3.2. Methodological framework

To achieve the objective of this study, this paper develops a two-phase sequential model to improve the effectiveness of UFP analysis in boosting TSFSCs by integrating the Analytic Network Process (ANP), a multi-criteria decision-making methodology, and Social Network Analysis (SNA). This model, differently from previous research conducted in UFP with ANP or SNA, integrates the two methods to provide a more comprehensive analysis of the importance of the territory in the design and implementation of UFPs. Thus, ANP is employed to determine the prioritisation of territorial factors that influence the development of TSFSCs. SNA is implemented to identify the changes implied by the UFPs within the TSFSC network (Fig. 1), since UFP was the first sub-factor prioritised with ANP. This methodological model is based on the knowledge and assessment of experts (actors in TSFSCs). In the rest of

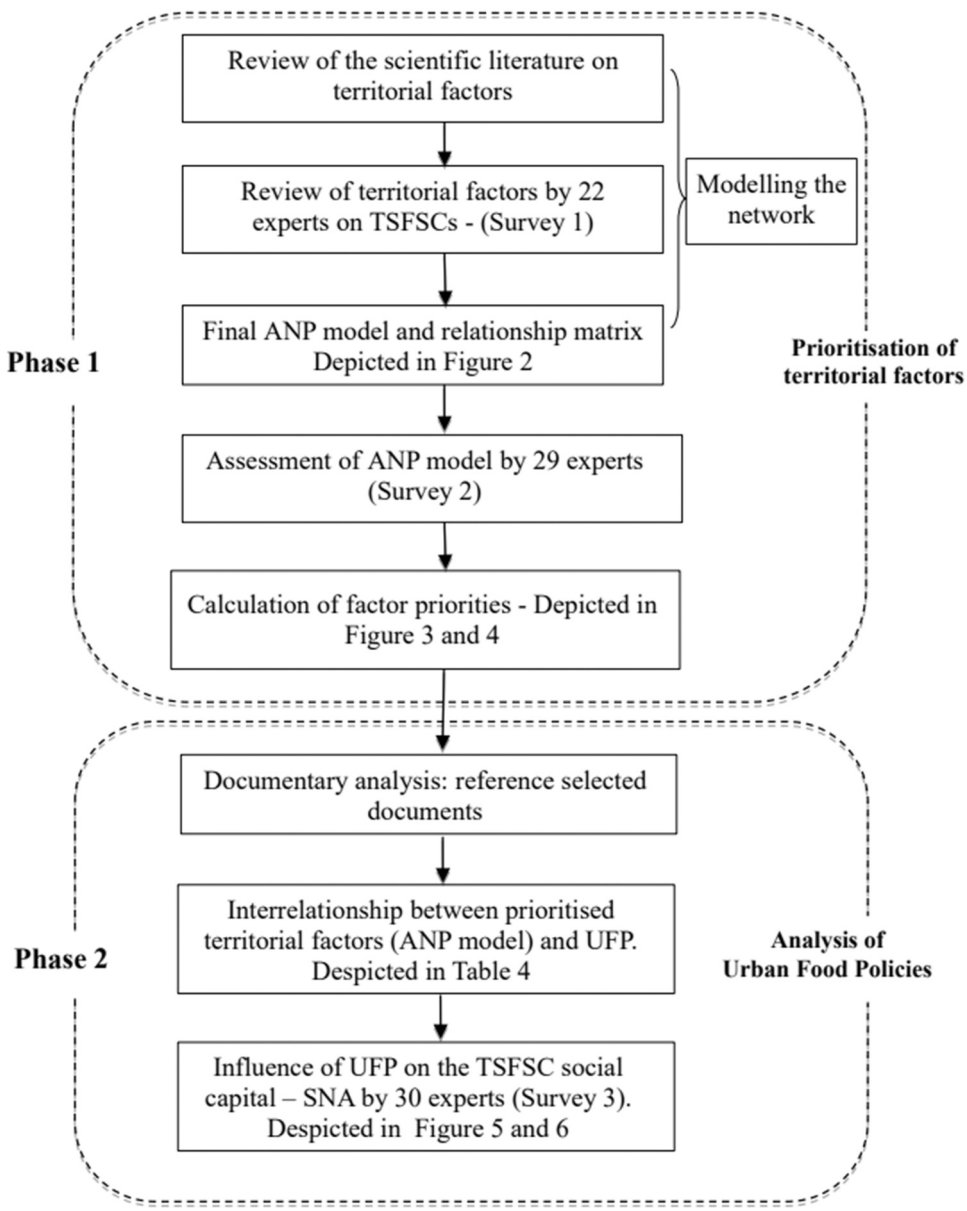

Fig. 1. Propose methodological model.

this section, a brief review of the ANP will be presented first, followed by the SNA.

3.2.1. Phase 1: Prioritisation of territorial factors through Analytic Network Process

Analytic Network Process (ANP) is a multi-criteria decision making methodology, which represents a decision problem as a network model composed by decision elements and clusters of elements, where every element can have an influence on itself or some, or on all the other elements of the system (Niemira and Saaty, 2004). It allows a better modelling of reality than simpler and linear models (Sánchez-Zamora et al., 2017). The use of ANP in scientific research has been increasing in recent years (Carmona-Torres et al., 2014; Niemira and Saaty, 2004; Rajesh, 2020; Saaty and Vargas, 2013). However, its application in agrifood networks has not been widespread. In particular, there is no research that focuses on the identification and prioritisation of territorial factors to boost AFN in cities based on ANP.

The full description of the theoretical and operational basis of the ANP method can be found in Saaty, 2001, but the following summarises the basic steps involved in applying the method, while explaining its application for this research:

1. Modelling the problem as a network: The design of the network model is one of the key points for the right problem solution (Saaty, 2001). The network basic units are the elements of the system under analysis, which are grouped into components or clusters. Subsequently, the relationships between these elements that compose the network are identified. In this research the clusters are 7 territorial factors, and the elements are the territorial sub-factors that compose each cluster (Fig. 2). The ANP model in this research has been defined from 1) review of the scientific literature and 2) 22 personal interviews with a structured questionnaire to different stakeholders of the agrifood sector and the TSFSCs in each city. The final model (Fig. 2) consists of 3 levels:

- Level I: It corresponds to the main objective, or goal, to be achieved in solving the problem. In this case, to prioritise the territorial factors influencing the development of TSFSCs.
- Level II: It consists of seven territorial factors: natural, human and intellectual, social, cultural, economic, productive, and political. These categories were selected based on Campagne and Pecqueur, 2014; and Ramos and Garrido, 2011.
- Level III: It corresponds to the sub-factors, which are the most specific items within each factor. They are defined in detail in Annex 1.

After defining the elements and the clusters of the network, the possible relationships between these elements need to be established. For this purpose, a relationship matrix has been defined, based on a deliberative process in conjunction with three experts (one from Colombia and two from Spain), selected directly for their authority in the field. In this matrix, the rows and columns consist of all network elements grouped in clusters, so that in the pairwise comparison they take the value 1 when the row element influences the column criterion, and 0 when there is no influence. Relationships can be internal, when both elements belong to the same cluster; or external, when the elements belong to different clusters. These relationships are shown in Fig. 2. Arrows relating the factor to its own sub-factors, and between its sub-factors, are internal; and arrows relating a factor to other factors, and their sub-factors, are external.

2. Assessment of ANP model: Once the network has been defined and the possible relationships between its elements established, the next step is to assess these relationships quantitatively. That is, to determine the relative priorities (or contributions) of the different

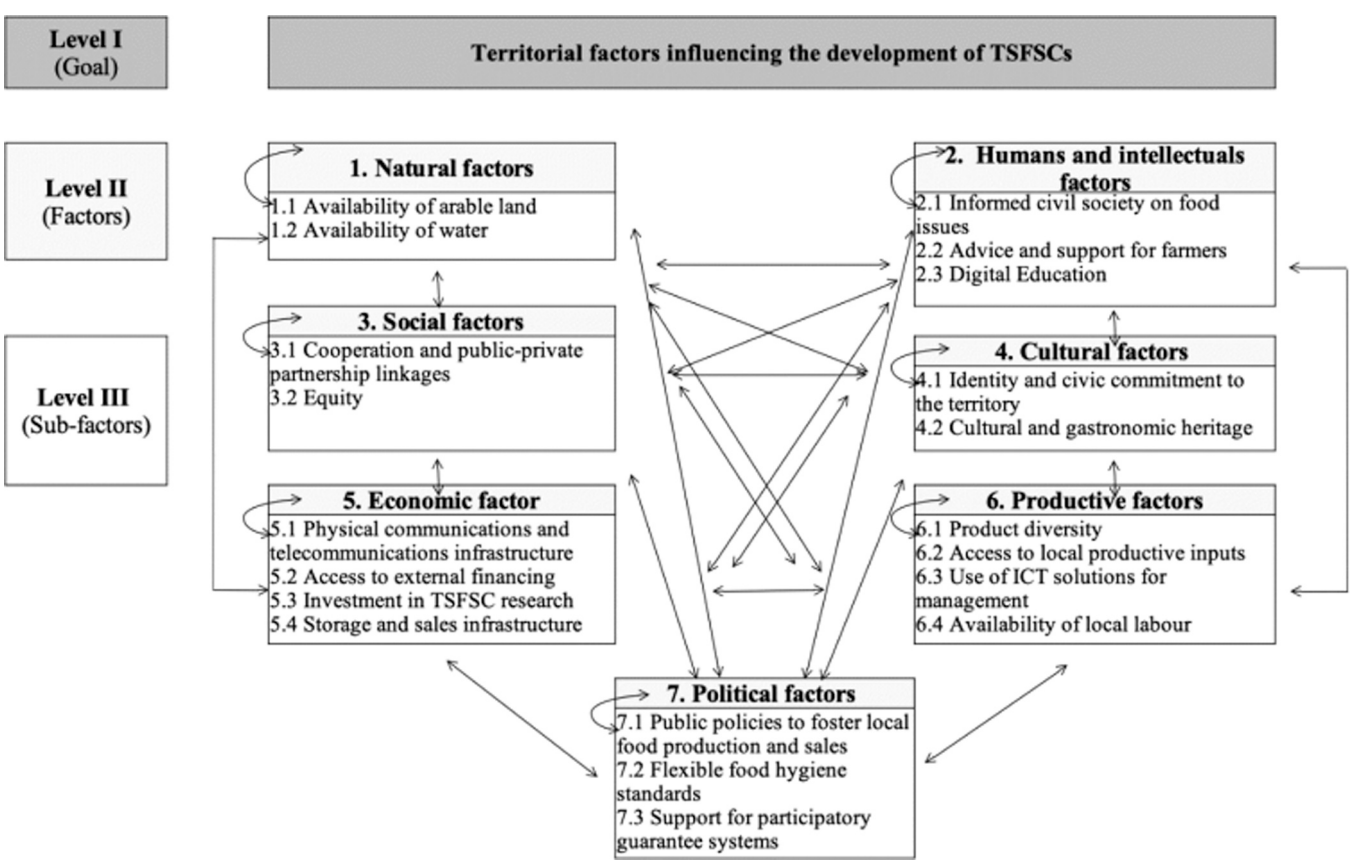

Fig. 2. ANP model for prioritisation of territorial factors influencing the development of TSFSCs.

elements (factors and sub-factors in our study) with respect to the element that dominates them, that is, on which they depend (Saaty and Takizawa, 1986). Due to the high number of elements to be compared in some cases (higher than 7 ± 2, as recommended in ANP), the evaluation of the priorities by the experts was based on the 'direct rating' method (Bottomley and Doyle, 2001; Calabrese et al., 2019; Larichev et al., 1995; Oliveira et al., 2018). Thus, the contribution of one element to another on which it depends has been obtained using a scale ranging from 1 (very weak contribution) to 9 (very strong contribution) (Carmona-Torres et al., 2014). This scale is linear, as indicated to the interviewees, where 9 is 9/1 times greater than 1, 9/2 times greater than 2, and so on. It is therefore to be expected that experts will think on this linear scale. This is a difference with the Likert scale where problems of interpretation of the scale may arise (Carifio and Perla, 2008; Jamieson, 2004; Pell, 2005). In order to obtain this information, a questionnaire was designed for TSFSC stakeholders in each city. Due to the requirement of in-depth technical knowledge and availability to complete long questionnaires, the number of experts to be consulted in the ANP methodology is reduced, and normally between 6 and 15 (Sánchez-Zamora et al., 2017; Villanueva Rodríguez et al., 2014). For this research, 29 experts were interviewed, 14 in Bogotá and 15 in Córdoba (some of them participated in the design model phase), categorised in different interest groups. The description of the experts can be seen in Table 1.

- Civil society: Representatives of civil society organizations, such as NGOs, social groups and funding institutions, among others.
- Academy: Persons belonging to higher education institutions and/or research centres.
- Market: TSFSC actors in the production, distribution or marketing stages.
- Public administration: Representatives of public administration agencies at the local and/or regional level.

3. Calculation of factor priorities: The first result of the completed questionnaires was an unweighted matrix for each expert interviewed (their columns do not add up to 1), with a total of 29 matrices. Each one of these unweighted matrices was multiplied by the cluster weighting or priority matrix to obtain the corresponding weighted supermatrix for each person (Saaty and Vargas, 2013). Once the individual weighted supermatrices were obtained, the local priorities were aggregated and an aggregated supermatrix was obtained. For this, the aggregation of individual priorities (AIP) was used, following the criteria of Forman and Peniwati, 1998, who recommend its use when stakeholders express their opinions individually. Subsequently, the aggregate supermatrix was then multiplied in parts, as proposed by Carmona-Torres et al., 2014, to obtain the priorities of each factor and sub-factor of the model. The results presented here refer, in general, to the global priorities, obtained as an average of the averages of the different groups of experts.

3.2.2. Phase 2: Analysis of Urban Food Policies

3.2.2.1. Documentary analysis and interrelationship between prioritised territorial factors and UFPs. Firstly, a critical analysis of UFPs and their relationship with territorial factors has been developed. Thus, the main documents that refer to UFPs in each city have been identified. The reference documents selected, because they were the most relevant to the research aims, are:

- Plenary agreement 292 of 2016 of the Córdoba City Council (Spain): Institutional declaration on the signing of the Milan Pact (MUFPP).
- Regulatory Decree 315 of 2006 of the Mayor's Office of Bogotá (Colombia): Master Plan for Food Supply and Security for the city of Bogotá (MPFSSB).

The period 2016–2020 has been selected as the framework for analysis, given that in 2016 the reference UFP is effective in Córdoba, although, in Bogotá, the reference UFP is effective since 2006. For the critical analysis of the UFPs, the policy cycle in each city has been identified. Although the phases of the policy cycle are agenda-setting, design, implementation and evaluation (Galli et al., 2020), this analysis focuses only on the design phase (normative documents: policy and programmes) and the implementation phase (execution of projects and concrete actions). Finally, the two selected phases of the UFP have been related to the territorial sub-factors they foster. Accordingly, the policy documents (regulations) and implementation documents (e.g., lists of projects and/or actions implemented) have been reviewed, and in each case the driven territorial sub-factors have been identified.

3.2.2.2. Influence of UFP on the TSFSC social capital. In order to identify the changes that the UFPs implied in TSFSCs, the social capital formed among TSFSC actors in the period 2016–2020 was selected as one of the variables that UFPs can influence. Social capital, according to Reina-Usuga et al., 2018, 2020 is an essential element for the emergence and growth of TSFSCs. Therefore, changes in some of the components of social capital may imply changes in TSFSCs. Social Network Analysis (SNA) techniques have been used for this analysis.

The SNA is a set of analysis techniques for investigating relationships between actors and social structures. SNA allows the study of empirical questions about governance and social capital in networks. Even relatively small groups of actors form complex network structures that can be discovered with SNA methods, but are otherwise difficult to detect, as the social structure is often a mixture of cohesive subgroups (Frank, 1996) linked by so-called weak or strong ties (Granovetter, 1973). Putman, 2001 states that social capital refers to the connections between individuals expressed through social networks and the norms of reciprocity and trust that emerge from them. Therefore, the SNA allows characterising and analysing such interactions between system actors, i.e., the social capital generated. The SNA aims to identify the type of network formed. This can be closed networks in which all nodes are connected or bridging networks in which the relationships are not as strong, but there are connecting nodes (Buciega and Esparcia, 2013). The academic literature refers to these two types of networks that generate social capital as bonding social capital and bridging social capital, respectively (Arnott et al., 2021; Cofré-Bravo et al., 2019; Dressel et al., 2020).

In the latter type of social capital, the bridge actors play an important role in the network, because they act as connectors between actors who have no direct connection. Thus, relations among diverse actors, and the betweenness centrality, indicate the configuration of a bridging social capital. In SNA, the betweenness centrality indicator is used to identify bridge actors (Borgatti et al., 1998; Putman, 2001). Thus, relations among diverse actors, and the betweenness centrality, indicate the configuration of a bridging social capital (Borgatti et al., 1998; Putman, 2001). SNA has been used to analyse social capital in different fields, such as LEADER rural development groups (Buciega and Esparcia, 2013), forest biodiversity conservation (Borg et al., 2015), reward-based crowdfunding (Eiteneyer et al., 2019), bioeconomy (Giurca and Metz, 2018), rural innovations (Cofré-Bravo et al., 2019; Kratzer and Ammering, 2019), adaptive capacity (Arnott et al., 2021; Dressel et al., 2020), among others.

Table 1
Experts consulted by interest group.

City	Civil society	Academy	Market	Public administration	Total interviews
Bogotá	4	3	4	3	14
Córdoba	4	4	4	3	15

In SNA, networks are formed by two basic elements: the actors (nodes) represented by points in the network and the relationships that are established (edges) represented by lines. The mathematical theory of graphs, matrixes and relational algebra is used in SNA (Scott, 2017). A large number of concepts have been developed to characterise, measure, and compare network structures and positions in networks. Table 2 shows some important indicators. Structure (cohesion) indicators are measures of the whole network, as opposed to node indicators which are individual measures.

In order to perform this analysis, an exploratory/descriptive research has been designed, based on the relationships created among TSFSC actors in each city. Most of the information used is primary, collected in two periods: 1) January-December 2016; and 2) January-March 2020 (Table 3). In the second period [January-March 2020], only the case of Cordoba has been analysed, because during this period of analysis, Córdoba joined MUFPP and created the Coordinating Board Milan Pact in Córdoba (CBMPC). The CBMPC led to significant changes in the relations between agri-food actors, which provided substantial elements for the comparison between the two periods. Nevertheless, during the second period in Bogotá, there were no changes in the UFPs, given that the ones that were created (RAPE and peace agreement) were not implemented, and consequently it is assumed that there were no changes in the relationships between food actors in Bogotá as a result of the implementation of UFPs. Although only the two periods in Córdoba are compared, this analysis is considered relevant, as it is the first time that changes in social relations and subsequent social capital as a consequence of the implementation of a UFP have been studied.

The nodes in this research are organisations linked to the TSFSCs of the two cities, these organisations can be both public and private. They are linked to production, distribution, sales or support organisations, such as public administration, NGOs or universities. The edges are the interactions between these organisations at the social level, i.e. with those who share information, organise events or participate in related activities on TSFSC issues (no monetary exchange).

In particular, the following techniques have been used to obtain primary information:

1. Personal interviews: 30 personal interviews have been conducted (20 in Bogotá and 10 in Córdoba) with TSFSC legal representatives or delegates of the initiatives. The snowball sampling technique was used, and sampling continued until data saturation (Bârsan-Pipu et al., 2002). Given that in Córdoba there are fewer organisations (not interactions), compared to Bogotá, and some delegates are involved in different organisations, the number of interviews is lower, because the saturation point was reached earlier. A structured questionnaire was used to identify interactions in the social sphere. In each interaction the intensity was rated by each interviewee, between 1 and 10 (1 = very low intensity, 10 = very high intensity of relationship).
2. Participant observation: 20 TSFSC initiatives were directly observed in key activities, such as a farmer market day (logistics, sales development, price setting, etc.), assembly meetings or other deliberative spaces, among others. The information was collected in a field notebook, also used to complement the information gathered in the interviews.

Table 3
Time periods for the SNA in Bogotá and Córdoba.

Geographical framework	2016	2020
Bogotá	Primary Information	Not applicable
Córdoba	Secondary information from the research by Reina-Usuga et al., 2018	Primary Information

The free open-source software Gephi 0.9 (https://gephi.org/) was used for the analysis of the information. Gephi allows the same network to be graphically represented in different ways using different layout algorithms. In this research, the Force Atlas layout was used, with a repulsive force of 200000, and a gravity of 30.0. This configuration allowed for: minimising crossings between edges, nodes not overlapping links that do not impinge on them and keeping a uniform edge length.

4. Results

4.1. Prioritisation of territorial factors and sub-factors

4.1.1. Priorities of factors

The seven territorial factors have different priorities (importance) with respect to the goal, on which they depend (Fig. 2), according to the average of all expert groups. Fig. 3 depicts the global priority of territorial factors in each city and indicates small differences in ranking. In descending order of importance are: 1) F5 Economic; 2) F2 Human and intellectual in Bogotá, and F7 Political in Córdoba; 3) Political F7 in Bogotá, and F2 Human and intellectual in Córdoba; 4) F3 Social; 5) F1 Natural; 6) F6 Productive; and F4 Cultural. It is worth to highlight that those factors related to the built resources of the territory (F5: physical infrastructure and F2: qualified human resources), as well as those referring to institutional arrangements (F7), have the highest priority for the development of TSFCS, over the specific territorial resources (F4).

On the other hand, despite the fact that both cities share a large part of the prioritisation ranking, an analysis of the global priority value shows that each city attributes greater importance to some factors, which responds precisely to the territorial variable. The most notable differences are found in F7, which is more important in Córdoba (0.0165), and F6, which is more important in Bogotá (0.133). The other factors have a short distance between the values.

4.1.2. Priorities of sub-factors

Fig. 4 depicts the global priority of territorial sub-factors in each city. The prioritisation of sub-factors shows that the two cities have similar general trends in the prioritisation when ranking them. However, the priorities in each city are different, which could be explained by territorial features. The most important sub-factor for the TSFSC, average for all expert groups, are: SF7.1 Public policies to foster local food production and sales, SF3.1 Cooperation and public-private partnership linkages, SF5.3 Investment in TSFSC research, SF2.2 Advice and support for farmers. Finally, SF6.4 Availability of local labour, and SF5.4 Storage and sales infrastructure, have a low importance in both cities. On the other hand, in terms of the differences in priority values, the most notable were found in: SF7.1 which is more important in Córdoba

Table 2
Main indicators in Social Network Analysis (SNA).

Structure of the network		Position of the nodes	
Average degree	Average number of steps needed to go from one node to any other	**Degree**	Number of connections of a node. It can be input or output
Network diameter	Greatest number of steps between any pair of nodes	**Proximity**	The node that's closest to all the nodes
Graph Density	The extent to which nodes are interconnected – lower density networks have fewer links between nodes. A complete network has all possible edges (relationships) and a density equal to 1	**Betweenness Centrality**	How central (on average) nodes in the network are. Average number of unique paths that pass through the nodes

Factors	Bogotá		Ranking	Córdoba		Ranking
F.1 Natural		0,1240	(5)		0,1305	(5)
F2. Human and intellectual		0,1778	(2)		0,1732	(3)
F3. Social		0,1334	(4)		0,1366	(4)
F4. Cultural		0,0951	(7)		0,0896	(7)
F5. Economic		0,1898	(1)		0,1871	(1)
F6. Productive		0,1184	(6)		0,1051	(6)
F7. Political		0,1614	(3)		0,1779	(2)

Fig. 3. Global priorities of territorial factor in Bogotá and Córdoba.

Sub-factors		Bogotá		Ranking	Córdoba		Ranking
SF1.1	Availability of arable land		0,0657	(5)		0,0628	(7)
SF1.2	Availability of water		0,0584	(7)		0,0677	(5)
SF2.1	Informed civil society on food issues		0,0646	(6)		0,0663	(6)
SF2.2	Advice and support for farmers		0,0796	(3)		0,0741	(4)
SF2.3	Digital Education		0,0337	(15)		0,0328	(14)
SF3.1	Cooperation and public-private partnership linkages		0,0834	(2)		0,0862	(2)
SF3.2	Equity		0,0500	(10)		0,0504	(9)
SF4.1	Identity and civic commitment to the territory		0,0442	(11)		0,0396	(11)
SF4.2	Cultural and gastronomic heritage		0,0508	(9)		0,0500	(10)
SF5.1	Physical infrastructure for communications and telecommunications		0,0570	(8)		0,0512	(8)
SF5.2	Access to external financing		0,0366	(13)		0,0360	(13)
SF5.3	Investment in TSFSC research		0,0790	(4)		0,0845	(3)
SF5.4	Storage and sales infrastructure		0,0173	(20)		0,0153	(20)
SF6.1	Product diversity		0,0363	(14)		0,0313	(15)
SF6.2	Access to local productive inputs		0,0278	(17)		0,0239	(17)
SF6.3	Use of ICT solutions for management		0,0316	(16)		0,0305	(16)
SF6.4	Availability of local labour		0,0227	(19)		0,0195	(19)
SF7.1	Public policies to foster local food production and sales		0,0990	(1)		0,1187	(1)
SF7.2	Flexible food hygiene standards		0,0255	(18)		0,0232	(18)
SF7.3	Support for participatory guarantee systems		0,0369	(12)		0,0360	(12)

Fig. 4. Global priorities of sub-factors in Bogotá and Córdoba.

(0.0197); SF1.2 which is more important in Córdoba (0.0093); and SF5.1 (0.0058) which is more important in Bogotá. The other sub-factors have a short distance between the values.

The similarities in the ranking of prioritisation of sub-factors in both cities have made it possible to define a typology of sub-factors. Each type of sub-factors has been conceptualised as having some common characteristic, as will be seen below. The types of sub-factors have been sorted according to their order of priorities and the range of priorities is recorded (based on Fig. 4). It is highlighted that the facilitating and driving sub-factors could be classified as means to achieve the other sub-factors, as they influence all other types of sub-factors. Hence, the other sub-factors (endogenous, access and operative results) could be classified as ends.

Thus, the typology of sub-factors is:

- Facilitating and driving sub-factors (priority range: 0.1187–0.0512): In this type, sub-factors have the highest priorities, and can be conceptualised as the means or vehicles to achieve other sub-factors, indicating that despite their importance they are not an aim in themselves. These are: SF7.1 Public policies to foster local food production and sales, SF3.1 Cooperation and public-private partnership linkages, SF5.3 Investment in TSFSC research, SF2.2 Advice and support for farmers, SF2.1 Informed civil society on food issues, and SF5.1 Physical infrastructure for communications and telecommunications. It is important to highlight the relevance of SF7.1 Public policies to foster local food production and sales, for both cities.
- Specific endogenous sub-factors (priority range: 0.0677–0.0395): Although the majority of sub-factors analysed are of a specific type, in this group they are included the own territorial resources, which can be socio-cultural or environmental. These are: SF1.1 Availability of arable land, SF1.2 Availability of water, SF4.2 Cultural and gastronomic heritage, SF3.2 Equity, and SF4.1 Identity and civic commitment to the territory.
- Access result sub-factors (priority range: 0.0359–0.0304): These are sub-factors that can be the consequence of the interaction or action of the two previous types of sub-factors, which favour access to specific resources required for the development of TSFSCs. These are: SF5.2 Access to external financing, SF7.3 Support for participatory guarantee systems, SF6.1 Product diversity, SF2.3 Digital education, and SF6.3 Use of ICT solutions for management.
- Operative result sub-factors (priority range: 0.0238–0.0152): This type of sub-factors can be catalogued as the purposes or outcomes pursued through the previous types, which directly affect the operation of TSFSCs. These are: SF6.2 Access to local productive inputs, SF7.2 Flexible food hygiene standards, SF6.4 Availability of local labour, and SF5.4 Storage and sales infrastructure.

Finally, based on the results of the prioritisation of territorial factors (described above), the first two prioritised sub-factors have been studied in depth, thus shaping the following two sections of results: 4.2 Urban

Food Policy (SF7.1 Public policies to foster local food production and sales) and 4.3 Influence of UFP on the TSFSC social capital (SF3.1 Cooperation and public-private partnership linkages).

4.2. Urban Food Policies

4.2.1. Interrelationship between UFPs and territorial factors

Table 4 is based on the prioritised categories of territorial sub-factors (explained in 4.1.2) and indicates which of these factors have been included in the current UFPs of Bogotá and Córdoba. Thus, in the design phase, the inclusion of facilitating and driving factors is highlighted in both cities, and in the case of the MPFSSB in Bogotá, some access and operational outcome factors. However, planning through programmes is not always carried out, as evidenced in the case of Córdoba. In the implementation phase, actions focused on the sub-factor SF3.1 'Cooperation and public-private partnership linkages' are highlighted in Córdoba. The Coordination Board of the Milan Pact (CBMPC) was created in this context. The CBMPC has had a great impact on the articulation of different types of food actors in Córdoba, despite not appearing in the political regulations, nor being part of a specific agency.

In Córdoba, the vast majority of the actions (without project) shown in Table 4 have been implemented by the CBMPC members and have focused mainly on the enabling and driving factors, with some specific cases on the access and operational outcome factors. On the other hand, the Bogotá UFP were focused on projects, at less in specific actions; these projects were linked mainly on enabling and driving factors as in Córdoba. However, in Bogotá the actions of SF3.1 Cooperation and public-private partnership did not have a direct influence on participatory governance mechanisms, as will be seen in the next section.

4.2.2. Influence of UFP on the TSFSC social capital

The SNA of stakeholders in the TSFSC is closely related to the second prioritised sub-factor SF3.1 Cooperation and public-private partnership linkages; and highlights the interaction of different types of actors, including public institutions, political or knowledge management institutions and private sector institutions, with a diversity of civil society organisations.

Fig. 5 represents the Bogotá network and Fig. 6a and 6b represent the Córdoba networks in the two time periods analysed. These figures points to that civil society institutions assume the greatest participation in the two cities. The involvement of production stands out, especially processed food, and to a lesser extent fresh food. In Bogotá, the participation of producers' associations, social collectives, and farmers' markets stands out. The participation of organisations linked to international cooperation (NGOs, foundations and cooperation agencies) also stands out, which together account for 27.28% of the organisations participating in this network. However, in Córdoba, the participation of small retailers and sales points, social collectives, NGOs, and public agencies predominates.

Table 5 shows some key indicators of the networks configured in the social sphere of both cities, and it shows the participation of a greater number of actors (nodes) in the social network in Córdoba compared to Bogotá. In terms of relationships (edges), Córdoba has more than twice as many relationships as Bogotá; this is reflected in the value of the diameter and density of the network, which is also twice as large in Córdoba. However, the density of the network is low in both cities (the ideal value is 1), indicating that not all actors are connected to each other. Finally, it is noteworthy to point out the variation in the value of the average degree in the Córdoba networks in both periods. Since between 2016 and 2020 it has decreased by almost half due to the number of actors has increased, but the average number of relationships of each actor has decreased, even though the overall number of relationships in the network has increased. The variation in the average degree can be explained by the creation of the CBMPC, as it facilitates the connection between nodes, shortening the path and decreasing the number of

Table 4
Connection between public policies and prioritised territorial factors (2016–2020).

Sub-factor		Córdoba (Spain)				Bogotá (Colombia)			
	Policy cycle[a]	*D*	*D*	*I*	*I*	*D*	*D*	*I*	*I*
		Policy	**Program**	**Project**	**Action**[b]	**Policy**	**Program**	**Project**	**Action**[b]
Facilitating and driving	SF7.1 Public policies to foster local food production and sales								
	SF3.1 Cooperation and public-private partnership linkages	X			X	X	X		X
	SF5.3 Investment in TSFSC research			X		X		+	
	SF2.2 Advice and support for farmers	X				X	X	+	
	SF2.1 Informed civil society on food issues	X		X		X	X	X	
	SF5.1 Physical infrastructure for communications and telecommunications					X		+	
Specific endogenous	SF1.1 Availability of arable land								
	SF1.2 Availability of water								
	SF4.2 Cultural and gastronomic heritage			X			X	X	
	SF3.2 Equity	X							
	SF4.1 Identity and civic engagement with the territory.			X					
Access result	SF5.2 Access to external finance				X				
	SF7.3 Support for participatory guarantee systems								
	SF6.1 Product diversity	X			X	X	X	+	
	SF2.3 Digital education								
	SF6.3 Use of ICT solutions for management					X			
Operative result	SF6.2 Access to local productive inputs				X	X		X	
	SF7.2 Flexible food hygiene standards								
	SF6.4 Availability of local labour								
	SF5.4 Storage and sales infrastructure				X*	X	X	+	X*

The first line (in grey), which corresponds to sub-factor SF7.1 Public policies to foster local food production and sales, is not filled in, as the table aims precisely to indicate which of the other sub-factors are included in the UFPs, and it would be redundant for it to be marked as such.
[a]*The table* includes the classification according to the first two phases of the policy cycle. *Design (D), Implementation (I).*
[b]Refers to actions implemented not linked to a specific project or programme, but in line with the UFPs
X Relationship between public policies and territorial factors
+Refers to projects implemented before 2016
*Refers to licences to farmer markets in public spaces

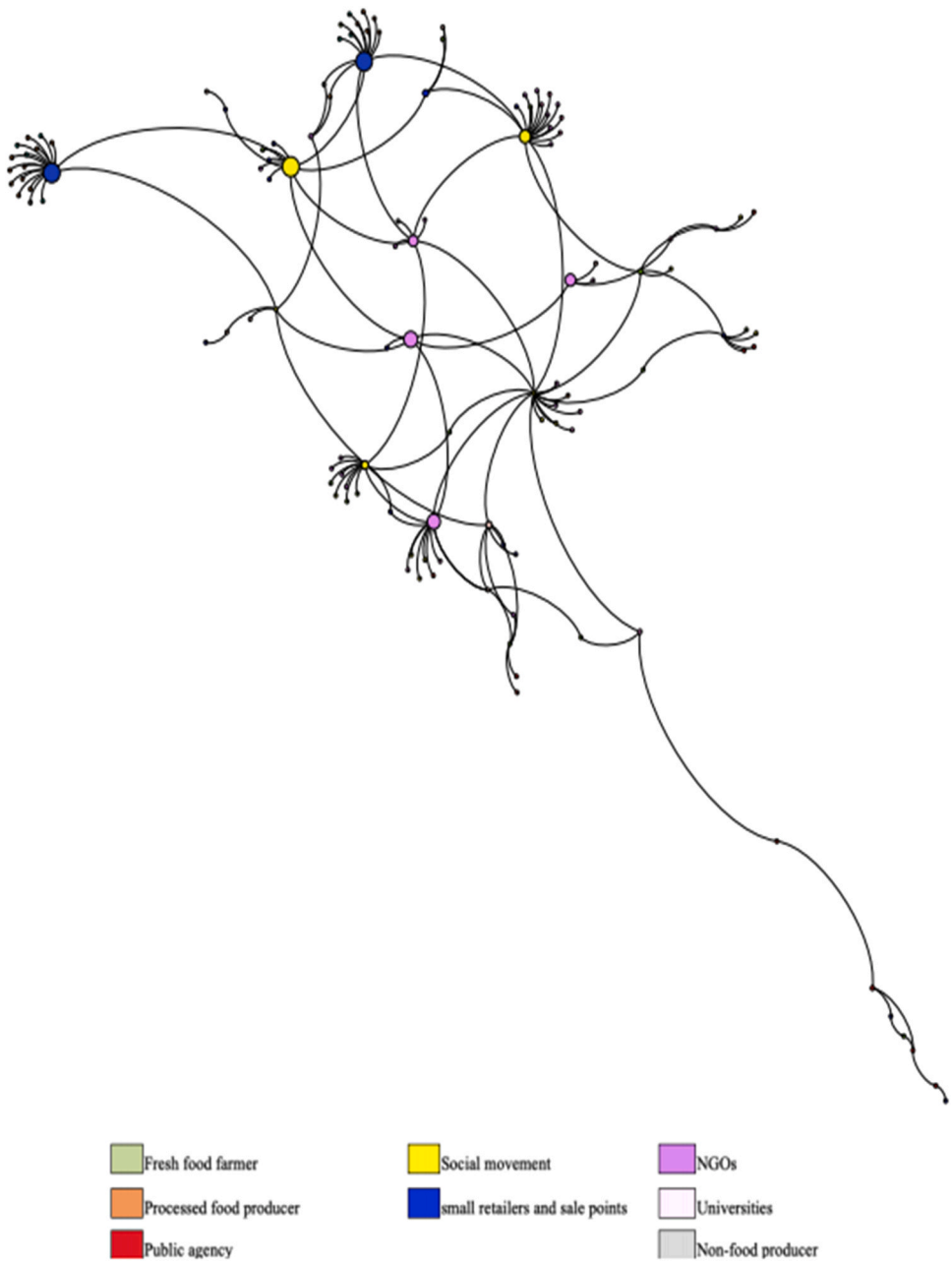

Fig. 5. Social network of the TSFSCs in Bogotá 2016.

interactions between them.

The main aim of the SNA in the two time periods is to identify whether there are significant changes in the network configurations because of the UFPs implementation. For this purpose, the number of actors, the density of the network and the bridge actors (identified by betweeness centrality and represented in the figures by the nodes with the largest size) are considered as a reference. In 2016, in Bogotá (Fig. 5) the bridge actors are NGOs, followed by a social movement (Red de Mercados Agroecológicos de Bogotá Región) and some agro-ecological farmer markets. In Córdoba (Fig. 6a), the bridge actors are of different types, with a predominance of social movement (an organic market stands out), small retailers, NGOs, universities, and a producer of processed foods.

In 2020, changes in the Bogotá network are not analysed because the UFP did not have variations with respect to 2016. However, the social network configuration does change in Córdoba (Fig. 6b). The main transformation is the emergence of the CBMPC in the social sphere. It is evident that the CBMPC has assumed a central intermediary position in the network, which in a way is positive, because this feature endows the node with a key position in the network structure, as it tends to influence communications between the other nodes. Central betweeness nodes, such as the CBMPC, are also called gatekeepers, because they seek to control the flow of information between communities. Thus, in a first step, the CBMPC has achieved the purpose of coordinating a diversity of food actors in the city.

Furthermore, the total number of actors has increased in Córdoba, although some have disappeared compared to 2016. According to participant observation, some of the organisations identified in 2016 mutated into other types of organisations in 2020, or some that aggregated many actors disappeared, but actors retained their participation on an individual level. It is also relevant that the average degree has decreased by almost half, which indicates that the intensity of the connection between nodes is low, so the role of the bridge nodes is key to avoid fragmentation in the network. Despite the decrease in this

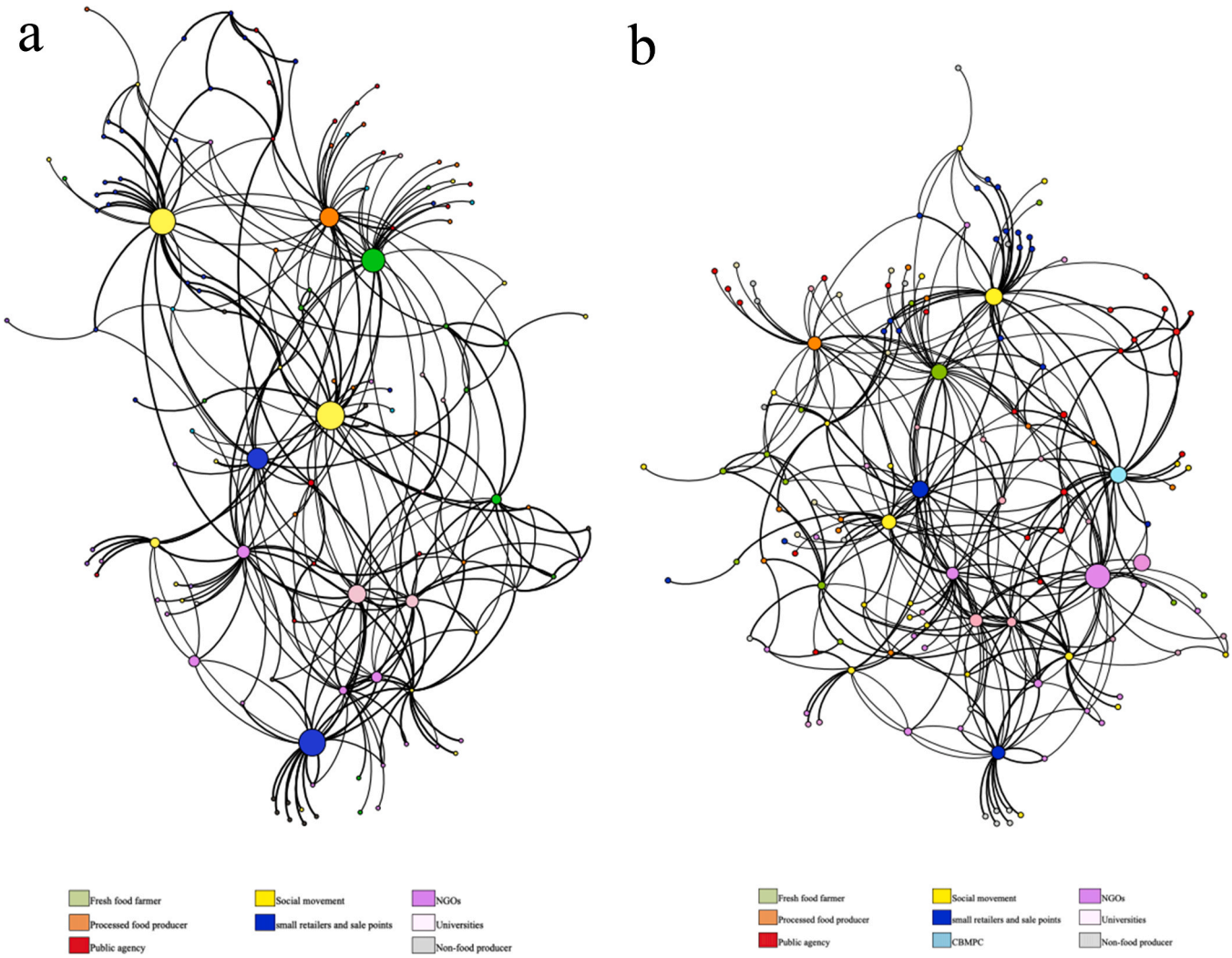

Fig. 6. (a) Social network of the TSFSC in Córdoba 2016. (b) TSFSC social network in Córdoba 2020.

Table 5
Indicators of the configuration of the social network.

	Nodes	Edges	Average degree	Network diameter	Graph density
Bogotá 2016	121	143	2.36	4	0.01
Córdoba 2016	130	290	*4.46*	8	*0.02*
Córdoba 2020	142	320	2.25	8	0.02

indicator, the variation in network density is minimal. Another point to note is that the size of bridge nodes has decreased, showing more of them, which could indicate a decrease in the concentration of power. However, these latter changes cannot be explained solely by the emergence of the CBMPC.

An approach to network structure is provided by two measures that provide information on the network as a whole, density and centralisation. Table 5 indicates that the density of relationships recorded in the network of the two cities is low. However, these are complete networks, as there are no disconnected actors. In terms of centralisation, it should be noted that there is no single central actor (centralised network), but that there is a set of actors at the centre of the network, structuring a decentralised network, indicating that power is substantially distributed throughout the structure.

On the other hand, according to participant observation, it was found that the creation of the CBMPC, through regular meetings, enabled the food stakeholders to dialogue among themselves and with the local public administration, in order to develop a food diagnosis of the city, and to design and implement joint actions to raise public awareness about food. In addition, two of the food actors, one a cooperative of fresh food and the other an association of small retailers, were granted a funding from the Andalusia Trade Integration Office, which allowed them to hire a person in charge of managing cohesion within them, and they also played an important role in the consolidation of the CBMPC.

Finally, according to the SNA and participant observation results, the interaction of different types of actors is stand out, between public and private institutions, with a diversity of civil society organisations. It also highlights the existence of bridge-actors that facilitate communication between unconnected actors. These two characteristics, the existence of relationships between actors with different characteristics and the existence of intermediation, indicate the configuration of bridging social capital. However, it is evident that the social capital that is formed in Bogotá is weak, and the departure of a bridge actor could imply the fragmentation of the network and the dissociation of peripheral actors. The opposite situation is evident in Córdoba, where, despite a low density, the actors show greater cohesion. This social capital created from the TSFSCs can be understood as a 'resource' which provides individuals and linked organisations with access to other types of capital. Here emerges the idea that behind the creation or support of these

networks there is intentionality from the actors involved. However, it can also be interpreted as an 'instrument', and not as an aim in itself. Social capital has the potential to generate collective action and boost processes such as TSFSCs, without forgetting the influence of the institutional framework, which can either strengthen or weaken it (Buciega and Esparcia, 2013). In this way, the social capital described should not be understood as a static and fixed element in Bogotá and Córdoba, but as a dynamic and transforming attribute in time, and based on food actor interactions.

5. Policy implications

5.1. Reflexions on current Urban Food Policies

Even though the analysis conducted in this research only included two of the phases of the policy cycle, the following reflections can be applied to the entire cycle. It is important to approach these implications with an interdependent approach among phases, which implies that the results of the previous phase affect the results of the following phases. Thus, starting in agenda and design phases, the territorial scale should be taken as the starting point and include a participatory coordination approach, which should be kept throughout the cycle. In the design and implementation phases, regulatory aspects must be taken into account, as well as the capacity for action and administrative and bureaucratic barriers. Finally, the evaluation phase should be considered as a tandem to each of the previous phases, i.e. each phase should include its own monitoring and evaluation mechanisms. Reflections for each of the phases are described below.

5.1.1. Agenda and design phases

5.1.1.1. Territorial scale. Cities have been estimated as an ideal territorial scale for UFPs. However, this term can be ambiguous if we compare the differences between Bogotá and Córdoba, not only in terms of geographical area and population density, but also in perceptions of locality, proximity, rurality, peasantry, and social demands such as food sovereignty (Sonnino et al., 2019). In addition, the definition of scale should be considered to overcome the lack of coordination between municipal, national (and European) food security aims and measures (Moragues-Faus et al., 2017). Therefore, the UFPs fostering TSFSCs should consider these elements and strengthen the deliberative spaces of reflexive governance, in order to identify problems and solutions in a participatory process, which may imply going beyond the geographical scale of cities and referring to food territories (FTs). Thus, UFP should focus on prioritised territorial factors in each city. For instance, when observing the territorial differences in the prioritisation of the sub-factors, it stands out that: SF7.1 Public policies to foster local food production and sales was the priority sub-factor, despite the fact that the difference in the global weight is the widest between the two cities. The global weight is higher in Córdoba (0.1187) than in Bogotá (0.099). This can be explained by the positive deployment and impact of public policies in rural Andalusia. As for Bogotá, the UFP has not had a major impact on the TSFSCs, although farmers' markets are part of the MPFSSB. The second sub-factor with the largest difference is SF1.2 Availability of water, which is more important in Córdoba (0.0677) than in Bogotá (0.0583). This can be explained by the fact that in Andalusia all river basins are affected by irregular rainfall, with marked seasonality, and the poor inter-annual distribution of rainfall, characteristic of the region's predominant Mediterranean climate. As a result, there is an unstable availability and recurrent shortage of water resources. The third sub-factor is SF5.1 Physical infrastructure for communications and telecommunications. In Bogotá, this sub-factor is highly ranked (0.0795) compared to Córdoba (0.0740). This could be explained by the potential benefit that this sub-factor represents for the development of the TSFSCs and also by the fact that small advances in road infrastructure, as well as the expansion of ICT coverage in rural areas, especially mobile coverage and internet access, have represented important resources for connectivity between the different food actors and the operation of the TSFSCs.

5.1.1.2. Participatory coordination mechanisms and food governance. The driving territorial sub-factor SF3.1 Cooperation and public-private partnership linkages, is one of the sub-factors that has been most boosted in the UFPs of Bogotá and Córdoba, and is evidenced by the social capital formed. This social capital brings about the design of new governance mechanisms based on spaces for deliberation, collective learning and it highlights the importance of the participation of territorial stakeholders in the planning and implementation of the UFPs. The CBMPC in Córdoba is a clear example of reflexive governance, and strengthening of the social capital, as has been evidenced in other cities that have established food policy councils (Brand et al., 2019; Filippini et al., 2019; Sonnino, 2016). However, as forementioned, it is critical to endow this mechanism with normative legitimacy, and to embody it in the UFP. On the other hand, in Bogotá the implementation of coordination and cooperation mechanisms between different types of food actors has only been addressed by civil society organisations, with the subsequent limitations to influence UFP and strengthen their social capital. Despite the fact that TSFSCs emerge from a strong social base (Reina-Usuga et al., 2020), the lack of legitimate coordination with local governments reduces the scope of action and, in the long term, the viability of the TSFSCs. Furthermore, this context could generate what the literature calls 'low hope scenarios' (Cohen-Chen and Van Zomeren, 2018) in which people do not believe that collective action can generate important changes in their reality, because, without hope, there is no basis for agency. In this sense, Sen, 1999 revisits the classic distinction between 'patient' and 'agent', to conceive the person not as a passive recipient of benefits or development aid but, on the contrary, as an active person who configures his own reality when looking for his/her destiny. These scenarios have already been evidenced in Colombian rural contexts, both civil society and local governments (Molina-Ochoa et al., 2019).

5.1.2. Design and implementation phases

5.1.2.1. Regulatory aspects. The above analysis showed that while many of the factors have been included in the planning phase, very few factors have been driven in the implementation phase. The main factor prioritised in the two cities is SF7.1 Public policies to foster local food production and sales. However, more than policies, projects and concrete actions have been implemented in this direction. If the UFPs are to be integrated by programmes and projects (Fernández and Schejtman, 2012), the UFP in Córdoba is a letter of intent, while the MPFSSB in Bogotá has a greater degree of normative concreteness. However, in practice, the actions carried out by CBMPC organisations, despite not being explicit in the normative policy, have had a significant impact on the mechanisms of participatory governance (Reina-Usuga et al., 2020). In contrast, the MPFSSB has failed to implement what it has proposed (Gutierréz, 2016.

To make progress in this area, cities need to move beyond the agenda-setting phase of the policy cycle and focus on the design and implementation phase. In the UFPs analysed, the design phase was not complete, as no programmes, projects, managers, timetables and budgets were defined. In addition, monitoring and control systems were not considered in the implementation phase.

5.1.2.2. Capacity for action and administrative and bureaucratic barriers. Historically, the process of policy integration in the agricultural and food sectors has been slow and to some extent contradictory and the UFPs in Bogotá and Córdoba are no exception. As indicated, TSFSCs emerge from collective actions. Therefore, they require time for their processes to shape into concrete actions which are sometimes outside

bureaucratic regulatory frameworks. But this same situation also emerges in public management, i.e. administrative times and processes do not respond in a timely manner to social demands. For example, in both cities organising the farmers' markets in public spaces has been a great obstacle for social groups, who have had to renovate their licences month by month, or even cancel markets. These licences depend on different administrative authorities at the local level, which has shown the low capability of coordinated reaction of the public administration, and the obsolescence of local administrative structures to the shift in the UFP paradigm.

5.2. Policy recommendations

The results allow us to suggest some relevant recommendations for the UFPs both at a general level and at a specific level. The general recommendations are described below, and the city-specific recommendations are described in Table 6.

a. The UFPs should include instrumental proposals that allow boosting and activating the factors that better tackle the problems in each city, thus contemplating the singularity of each territory.

Table 6
Recommendation of actions for the UFP in Bogotá and Córdoba.

Lines	Actions	
	Bogotá	**Córdoba**
Governance and social participation		
Food policy coordination structures, based on the participation of actors from different spheres and levels	Promote an Intersectoral Commission on Food and Nutrition Security, allowing the participation of civil society actors and social movements	Enhancing the CBMPC
Foster a city-region approach	Activate the RAPE and define specific functions to formulate and implement UFPs	Progress could be made on the basis of Rural Development Groups linking Córdoba as a city and a province
Promote the linking and/or creation of networks in different areas and levels, with an operational and practical vision	For example, adherence to the Milan Pact, as the city of Medellin (Colombia) has already done.	Strengthening community and commercial networks
Territorial factors		
Facilitating and driving sub-factors	- Public food procurement - Spaces for farmers' markets - Paving of tertiary level roads (urban and rural) - Collective aqueducts	- Public food procurement - Spaces for farmers' markets - Regulations for direct sales
Access result sub-factors	- Specific credit lines for small productions adapted to crop cycles and supported with public funds. - Subsidies adapted to the characteristics of family farming and rural territories. - Participatory Guarantee Systems	- Participatory Guarantee Systems
Operative result sub-factors	- Collective management of resources: i) Collective means of transportation for inputs and/or final product; and ii) Collective purchasing of production inputs - Spaces for storage and logistical centres	- Spaces for storage and logistical centres

b. The UFPs should be endowed with a regulatory framework, inevitably leading to the design phase. They should adopt an integrative and flexible approach, and include progress findings and corrective measures.
c. There is a need to move from the fostering of the facilitating and driving sub-factors, to the access and operative result sub-factors, which can resolve the impeding obstacles to the development of TSFSCs.
d. The development of food systems depends on factors that are linked to the scope of different public policies, which suggests the need for coordination and complementarity in the actions of public administration and civil society.
e. Innovation in UFPs requires paradigm shifts in public administration, both in instrumental management and in the establishment of coherence measures in their implementation, such as the use of public procurement as a tool for the support and sustainability of TSFSCs from the public administration.

6. Conclusions

The results indicate that factors such as public policies, cooperation links, advice and support for production, an informed civil society (on food issues), and the physical infrastructure of communications and telecommunications are key means to promote the emergence and development of TSFSCs. However, each city takes action according to its endogenous factors (type of production, population variables, armed conflict, natural resources, etc.). Results also highlight a key element for food policies to have an influence, and that is the instrumental design of such policies in a territorial approach. Policies for the food territory should consider the territorial singularities and particularities, which often transcend the local sphere. Thus, UFPs deserve a territorial approach, with various aims and measures (flexibility), or at least different priorities. In addition, it should be noted that the factors that have driven the emergence of TSFSCs are not necessarily the same as those that have enabled their stability (development).

On the other hand, the second prioritised territorial sub-factor is strongly related to the promotion of participatory coordination mechanisms (facilitating and driving sub-factors) that involve civil society, and therefore represents an important element to include in UFPs, as it favours the strengthening of collective action and social capital in cities. Achieving sustainable food systems requires participatory and reflexive governance processes. This point is relevant given that the results obtained show that the TSFSCs are supported by a strong social base, measured in this research by social networks, in which organisations of various kinds participate, with a predominance of civil society organisations that are connected, but with a low density of relationships, which favours a lack of centrality of power. However, there are bridge-actors that connect isolated groups of actors and highlight their role as network cohesors. In Bogotá, the role of NGOs and international cooperation agencies as cohesive actors in the network stands out, which could call into question the degree of empowerment of the local community. In Córdoba, the cohesive role of social collectives and small retailers stands out, which could question the role of consumers in the TSFSCs.

Thus, the UFP should be endowed with a regulatory framework that legitimises its programmes and projects, as well as its deliberative processes and decision-making mechanisms. These processes must be supported by a revitalisation of local public administration, which improves responsiveness and enhances inter-administrative public coordination. In this sense, the UFPs should also be equipped with mechanisms to overcome the electoral situation in the municipalities where they are implemented. It also highlights the promotion of different types of territorial sub-factors to respond to the operational needs of the TSFSC. While the first steps of the analysed UFPs focus on enabler and driver sub-factors, broadening their scope to access and operational outcome sub-factors could provide more tangible and short-term results for the

food territory.

Consequently, from the results obtained it can be concluded that the TSFSCs are related to the dynamics of the territory and not only to the characteristics of the products that flow through them, thus emerging as an endogenous territorial strategy, and becoming an element for sustainability and food governance. Therefore, the existence of territorial factors that are associated with the development of supply chains, such as UFPs and participatory coordination mechanisms, indicate the potential transformative role of TSFSCs in food governance that enables the achievement of sustainable food systems Although the results should not be generalised beyond the cities analysed, the proposed methodological model can be extrapolated to the analysis of other territories. The results achieved should not be understood as a fixed and static priority in Bogotá and Córdoba, but as a dynamic attribute that changes over time and depends on the interaction of the different actors involved in the territory's food system.

Finally, this study proposes a multi-criteria ANP model combined with SNA topological measurements to analyse the drive of UFP to TSFSCs. Innovatively combining these two non-traditional methodologies (ANP and SNA) allows for the design and deployment of programmes and projects, within UFP frameworks, that are more successful than many of those currently in place. Reducing the failure of UFPs in boosting TSFSCs will strengthen the opportunities to achieve more sustainable food systems. As shown through the empirical study, the proposed model can incorporate complex decision-making processes as a prioritisation of territorial factors to foster TSFSCs with social network metrics in order to track actual changes in food networks as a result of policies. This work only evaluated a social network at one point in time (e.g. when the survey was taken), but if social networks were evaluated over time, different conclusions relative to the importance of organizational relationships may unfold. Within the field of food policy, identifying the possible interactions between the ANP results and SNA criteria could contribute to the development of a more comprehensive model of ex ante and ex post assessment of UFPs.

Annex 1. Territorial factors and sub-factors influencing the development of TSFSC

Factor	Sub-factors		Definition	Unit of measure / Indicator
F1. Nature	**SF1.1**	Availability of arable land	Availability and access to arable land in or near the territory	Hectares
	SF1.2	Availability of water	Availability and access to water for agricultural use	Nº of water sources
F2. Human and intellectual	**SF2.1**	Informed civil society on food issues	Social groups in the field of consumption and production with knowledge of food issues, health and food quality	% of population informed (over total)
	SF2.2	Advice and support for farmers	Training of farmers in production processes (ecological, transition), post-harvest management, adding value to customer service, marketing and commercialization, negotiation.	% of farmers advised
	SF2.3	Digital Education	Knowledge and use of new technologies and the Internet (the area of consumption and production)	% of population educated (over total)
F3. Social	**SF3.1**	Cooperation and public-private partnership linkages	Creation and strengthening of support networks for lobbying, information and knowledge sharing, including public-private partnerships.	Nº of support networks
	SF3.2	Equity	Encourage the participation of women and young people in the AFN	% of women and young people
F4. Cultural	**SF4.1**	Identity and civic commitment to the territory	Sense of belonging to the territory	Personal perception
	SF4.2	Cultural and gastronomic heritage	Social and cultural value of food and places of purchase	Personal perception
F5. Economic	**SF5.1**	Physical communications and telecommunications infrastructure	Public investment in land transport networks and telecommunication network infrastructure	
	SF5.2	Access to external financing	Availability of credits and financing to productive projects	Nº of credits available
	SF5.3	Investment in TSFSC research	Support from academia in the fostering and strengthening of TSFSC	Nº of research projects
	SF5.4	Storage and sales infrastructure	Availability of places to sell and/or store food in the AFN	Nº of infrastructure
F6. Productive	**SF6.1**	Product diversity	Supply of diverse foods (vegetables, fruits, nuts, mushrooms, cheese, bread)	Nº of product ranges
	SF6.2	Access to local productive inputs	Availability of resources such as native seeds and other required inputs	
	SF6.3	Use of ICT solutions for management	Software for order management and delivery	
	SF6.4	Availability of local labour	Available and trained workers for agricultural work	Nº of rural workers
F7. Political	**SF7.1**	Public policies to foster local food production and sales	Public programmes and projects that foster the production and consumption of local food with environmental and fair practices, as well as local sales.	Nº of policies
	SF7.2	Flexible food hygiene standards	Food hygiene standards adapted for small-scale farms and new enterprises	Nº food hygiene standards
	SF7.3	Support for participatory guarantee systems	Support for participatory guarantee systems (PGS) proposed by civil society	Nº PGS

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